Fiscal Year:	FY 2017	**Task Last Updated:**	FY 06/19/2017
PI Name:	Van Heugten, Anthony		
Project Title:	e-Vision Electronic Adjustable Power Eyeglasses for Space and Earth		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Smart Medical Systems and Technology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	tonyv@evisionoptics.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	941-504-5722
Organization Name:	e-Vision Smart Optics, Inc.		
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PI Address 2:			
PI Web Page:			
City:	Sarasota	**State:**	FL
Zip Code:	34243	**Congressional District:**	16
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI-RFA-SMARTCAP
Start Date:	12/01/2015	**End Date:**	01/31/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 1/31/2017 (originally 11/30/2016) per NSBRI (Ed., 8/19/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-SMST00013		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	(1) Original project aims/objectives. This Project will consist of three phases – During Phase I of the Project, e-Vision will produce a prototype electronic lens with one adjustable optical zone. The prototype lens will be recharged and controlled via a wire to a control box. Wireless power and control will be included in Phase II of the Project. Optical power and switching speed of the Phase I prototype will be optimized to determine the design parameters for the Phase II portion of the project (currently unfunded), during which the electronics will be embedded inside of the lens. Once the optimum optical power and switching speed is determined from a single lens element, then the design can be finalized to include the required number of liquid crystal layers to achieve the required power. (For example, if a single lens layer can produce 1.5 Diopters of optical power, and 3 Diopters is required, then two layers will be required to be included in the Phase II design.) Phase III of the project will be to bring all aspects of the development efforts together to produce flight-ready glasses. (2) Key findings. We pursued two design strategies, Concentric Electrodes (type A), and linear electrodes (type B). Type A design strategy proved to be most successful, providing adequate optical image quality, switching speed and optical power. The optimum balance of optic size that delivers the best combination of good optical quality and sufficient optical power is currently 30 mm diameter. However, optical modeling indicates that 42 mm diameter is highly likely to achieve. The lens's optical power can indeed be tuned in an analog fashion and is not limited to simple on/off switching. The optimum optical power per layer of liquid crystal is 0.8 Diopters and the transmission data (i.e., light loss per layer), indicates that four lens cells can be stacked resulting in an optical power range of zero to 3.2 Diopters yet still transmitting 70% or more of the light. The lenses can be adjusted across 90% of their optical power range in one second or less. The lenses can be used for two purposes, the first purpose is to correct for a hyperopic shift and the second purpose is to provide ADD optical power for near vision magnification (so long as the user's magnitude of hyperopic shift has not reached the optical power limit of 3.2 Diopters). (3) Impact of key findings on hypotheses, technology requirements, objectives, and specific aims of the original proposal. The Key Findings support the assumption that electronically-adjustable lenses may be used to correct for hyperopic shifts that may occur during long-duration microgravity missions. Also confirmed is that the lenses have the added feature of providing additional optical power for near vision if not being used for Hyperopia correction. The unknown still is the diameter limit of the optical lens. The current 30 mm diameter may need to be increased to at least 42 mm (which is an acceptable diameter in the form of "John Lennon Glasses"). (4) Proposed research plan for the coming year. Increase the size of the optics. Integrate into wearable frames connected with a wire to a controller. Change optical power using a remote controller in either a smart-phone or a smart-watch. (We have not yet submitted a funding request for this work, but plan to.)
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Having adjustable prescription glasses will help people in numerous ways: - Astronauts experience unpredictable vision changes during long periods of exposure to microgravity. Having the ability to change the optical power of the glasses will allow the astronaut to make vision adjustments as needed during the course of the mission rather than carrying a large assortment of various lenses with them into space. - Presbyopia (loss of ability to change eye focus). Everybody eventually suffers from this. Glasses that can adjust prescription "on the fly" will provide a better optical result than those that have two or more optical powers permanently embedded in the glasses. When not being required, the near-vision prescription can be switched off allowing a larger area of the lens to be used for distance vision, removing the defocused zone from the lens, providing a better, less obstructed view. Also, with multiple focus points, an optimum large-zone-correction will be provided for not only near distance needs such as reading, but mid distance needs such as mobile phone and computer use and interacting with daily life devices such as ATMs, automobile controls, etc. - Changing visual prescription costs can be reduced. Currently, requiring a different prescription results in having a new pair of glasses made, which is an economic burden. As a result, even with insurance, changes are generally limited to a new pair every two years. If the glasses' optical power can be changed as needed, not only could people's vision be corrected to optimum as-needed regardless of time intervals, but the life of the glasses could be prolonged by reprogramming rather than replacing.
Task Progress:	Percent completion: Model and Fabricate lens designs A and B -- 100% ; Choose design A or B -- 100% ; Modify Lens Design Software from Experience Gained -- ongoing, continual process ; Model and Fabricate Lens Design Chosen -- 100% ; Design / Build Spectacle Frames -- 50% ; Design / Build Control electronics -- 100% ; Integrate and Test Frames / Lenses / Controllers -- 0%. Note: Our original goal / plan was to have the lenses integrated into a pair of frames for users to wear and try, but we did not complete that milestone in this period. Progress was made in that task, but not completed.
Bibliography Type:	Description: (Last Updated:)